Lecture 7

# Braiding: hexagons, Yang–Baxter and topological spin

**Objectives.** *Lectures 5 and 6 built fusion categories, with their particles and associators. In three dimensions, exchanging two identical particles twice is the identity, and exchange statistics are those of the symmetric group: bosons and fermions. In a plane the two exchanges are not the same, and the symmetric group is replaced by the braid group. This lecture adds a braiding to a monoidal category, derives the braid relation from its axioms, writes the axioms as equations in F- and R-symbols, and solves them for Fibonacci. Closing a braided strand gives the topological spin.*

By the end of this lecture you will be able to:

- State the braid group and why it replaces the symmetric group in a plane.
- Define a braiding, prove the braid relation from the hexagons, and tell the braid form of the Yang–Baxter equation from *Integrability*'s form.
- Write the hexagon equations in F- and R-symbols and solve them for Fibonacci.
- Compute topological spins, and find the braidings of the pointed categories on $\mathbb{Z}_2$.

## 7.1 $B_n$ versus $S_n$

The *braid group* $B_n$ has generators $\sigma_1, \dots, \sigma_{n-1}$, with

$$\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1},$$
$$\sigma_i\sigma_j = \sigma_j\sigma_i \quad (|i-j| \geq 2).$$

Adding $\sigma_i^2 = 1$ gives the symmetric group $S_n$. The generator $\sigma_i$ exchanges particles $i$ and $i+1$, passing the left one over:

$$\sigma_i = \quad \begin{matrix} \vert & \cdots & \diagdown & & \cdots & \vert \\ 1 & & i & i{+}1 & & n \end{matrix}$$

A point of a configuration space specifies all particle positions at once. A path in that space is a possible motion without collisions. A closed path returns to the same unordered configuration, although the individual particles may have exchanged places. Choose one starting configuration. The fundamental group collects closed paths based there, identifying paths that can be continuously deformed into one another without collisions and with that configuration fixed. Its product follows one path by another. For these connected spaces, simply connected means that every closed path can shrink to a point. The codimension of an excluded set is the number of independent local conditions defining it. Requiring two positions to agree imposes one condition for each spatial coordinate. These meanings explain the geometric statement below; its planar case is cited.

**Theorem 7.1** (Configuration spaces). *The fundamental group of the space of $n$ unordered distinct points in $\mathbb{R}^2$ is $B_n$ · cited (Kassel–Turaev §1.4.3); in $\mathbb{R}^d$ with $d \geq 3$ it is $S_n$ · sketched.*

For $d \geq 3$: ordered configurations form $\mathbb{R}^{dn}$ with the diagonals $x_i = x_j$ removed, each of codimension $d \geq 3$, and removing subspaces of codimension at least 3 leaves a space simply connected. The unordered space is its quotient by the free action of $S_n$, so its fundamental group is $S_n$. In the plane the diagonals have codimension 2, like a point in a plane, and a loop taking one particle around another does not shrink. Quantum mechanics of identical particles in a plane therefore allows unitary representations of $B_n$, not only representations of $S_n$: *anyons* (*Symmetries* §15.4).

## 7.2 Braided categories and hexagons

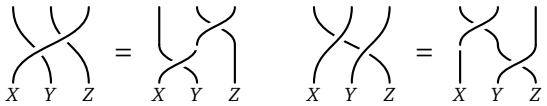
As in Lecture 4, we work as if the category were strict.

**Definition 7.2** (Braiding). A *braiding* on a monoidal category is a natural isomorphism $c_{X,Y} : X \otimes Y \to Y \otimes X$, drawn with the strand from bottom left to top right over the other, such that the *hexagons* hold:

$$c_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes c_{X,Z}) \circ (c_{X,Y} \otimes \mathrm{id}_Z),$$
$$c_{X\otimes Y,Z} = (c_{X,Z} \otimes \mathrm{id}_Y) \circ (\mathrm{id}_X \otimes c_{Y,Z}).$$

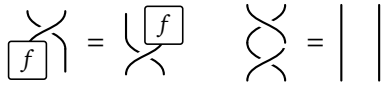
It is *symmetric* if $c_{Y,X} \circ c_{X,Y} = \mathrm{id}_{X\otimes Y}$ for all $X, Y$.

With the associators put back, each hexagon has six sides, hence the name (EGNO Def. 8.1.1). In pictures, the first says that a strand crossing over a pair crosses each in turn, and the second that a pair crossing over a strand does so one at a time:



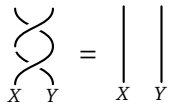

(7.1)

Naturality says that boxes slide through crossings, and invertibility that a crossing followed by its inverse is no crossing at all (Reidemeister II):




(7.2)

The *double braiding* $c_{Y,X} \circ c_{X,Y}$ takes one strand once around the other. A braiding is symmetric exactly when the double braiding is always trivial:




(7.3)

and then over- and under-crossings agree, and are drawn as a plain crossing.

**Example 7.3** (Braidings).

- Vect and Rep($G$) with the flip $v \otimes w \mapsto w \otimes v$, which is an intertwiner and natural: symmetric.

- *Super vector spaces* sVec: $\mathrm{Vec}_G$ on $\mathbb{Z}_2 = \{1, s\}$ with $c_{s,s} = -1$ (EGNO Ex. 8.2.2). It is symmetric, since $(-1)^2 = 1$, and the odd line is a fermion: exchanging two of them gives a sign.
- The same category with $c_{s,s} = +1$: symmetric again, but a different braiding, not equivalent to the first as a braided category: an equivalence fixes $s$, so it would preserve $c_{s,s} = \pm 1$. A braiding is structure, not a property.

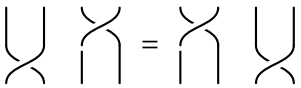
## 7.3 Yang–Baxter is Reidemeister III

**Theorem 7.4** (The braid relation · *proved given coherence*)**.** *In a braided category, for all $X, Y, Z$,*

$$\begin{aligned}(c_{Y,Z} \otimes \mathrm{id}_X)(\mathrm{id}_Y \otimes c_{X,Z}) \\ \circ (c_{X,Y} \otimes \mathrm{id}_Z) \\ = (\mathrm{id}_Z \otimes c_{X,Y})(c_{X,Z} \otimes \mathrm{id}_Y) \\ \circ (\mathrm{id}_X \otimes c_{Y,Z}).\end{aligned}$$

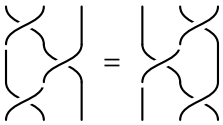
*So $\sigma_i \mapsto \mathrm{id} \otimes c_{X,X} \otimes \mathrm{id}$, in positions $i, i+1$, is a representation of $B_n$ on $X^{\otimes n}$.*

*Proof.* By the first hexagon the last two factors on the left are $c_{X,Y\otimes Z}$. Naturality of $c_{X,-}$ applied to $c_{Y,Z} : Y \otimes Z \to Z \otimes Y$ gives $(c_{Y,Z} \otimes \mathrm{id}_X)\, c_{X,Y\otimes Z} = c_{X,Z\otimes Y}\, (\mathrm{id}_X \otimes c_{Y,Z})$, and the first hexagon again turns $c_{X,Z\otimes Y}$ into the first two factors on the right. The far commutation $\sigma_i \sigma_j = \sigma_j \sigma_i$ is the interchange law (Theorem 3.4):



■

In pictures this is Reidemeister III: a strand slides over a crossing.



(7.4)

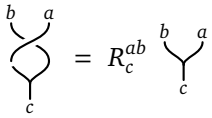
Let $P$ be the flip, $P(v \otimes w) = w \otimes v$. For a map $R$ on $V \otimes V$, a leg subscript specifies which ordered factors of $V^{\otimes 3}$ it acts on, leaving the other fixed:

$$\begin{aligned}R_{12} = R \otimes \mathrm{id}, \qquad R_{23} = \mathrm{id} \otimes R, \\ R_{13} = P_{23}(R \otimes \mathrm{id})P_{23}.\end{aligned}$$

*Integrability*'s Yang–Baxter equation carries spectral parameters (its eq. (14.8)). For a constant matrix $R$ on $V \otimes V$ it reads $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$, and then $R$ satisfies it exactly when $c = P \circ R$ satisfies the braid form $c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}$, which Kassel (Thm. XIII.1.3) and EGNO (Prop. 8.1.10) call the Yang–Baxter equation. Three things are called R, and must be kept apart: *Integrability*'s $R(\lambda)$, the braiding $c = P \circ R$, and the R-symbols of the next section.

## 7.4 R-symbols; Fibonacci solved

In a multiplicity-free fusion category with the vertices of Lecture 5, the braiding maps the line $\mathrm{Hom}(c, a\otimes b)$ to the line $\mathrm{Hom}(c, b\otimes a)$, so it acts on vertices by numbers, the *R-symbols*:




(7.5)

that is $c_{a,b}\circ\psi_c^{ab} = R_c^{ab}\,\psi_c^{ba}$, with $R_c^{ab}=1$ when $a$ or $b$ is 1: the first hexagon with $Y=Z=1$ gives $c_{X,1}=c_{X,1}^2$, so $c_{X,1}=\mathrm{id}$, and likewise $c_{1,X}=\mathrm{id}$. In a unitary category $\left|R_c^{ab}\right|=1$. A gauge transformation (Theorem 6.4) changes $R_c^{ab}$ by $u_c^{ba}/u_c^{ab}$, so $R_c^{aa}$ is a gauge invariant.

**Theorem 7.5** (The hexagon equations · *proved given coherence*). *The F- and R-symbols of a braided multiplicity-free fusion category satisfy, for all labels,*

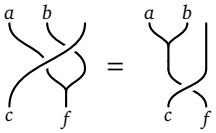
$$R_e^{ca}\,[F_d^{acb}]_{eg}\,R_g^{cb} = \sum_f [F_d^{cab}]_{ef}\,R_d^{cf}\,[F_d^{abc}]_{fg},$$

*and, from the second hexagon,*

$$(R_e^{ac})^{-1}[F_d^{acb}]_{eg}(R_g^{bc})^{-1} = \sum_f [F_d^{cab}]_{ef}\,(R_d^{fc})^{-1}[F_d^{abc}]_{fg}$$

*(Bonderson (2.57)–(2.58)). Conversely, F- and R-symbols satisfying the pentagon and both families define a braiding* · sketched.

*Proof.* The step that does the work is that a strand crosses a vertex as it crosses the two strands leaving it, by naturality and the first hexagon:




(7.6)

Apply both sides of the first hexagon, with $X=c$, $Y=a$, $Z=b$, to the tree in which $a$ and $c$ fuse first, through $e$, to total charge $d$. On the right, $c$ crosses $a$, which multiplies by $R_e^{ca}$; an F-move brings $c$ and $b$ together, through $g$; $c$ crosses $b$, which gives $R_g^{cb}$. On the left, an F-move brings $a,b$ together, through $f$; $c$ crosses the pair $f$ at once, giving $R_d^{cf}$; a last F-move returns to the basis of trees fusing $b$ and $c$ first. Comparing coefficients of the basis trees gives the equation. The second family comes the same way from the second hexagon, written for the inverse braiding: $c_{b,a}^{-1}$ acts on $\psi_c^{ab}$ by $(R_c^{ba})^{-1}$, so inverting a crossing also exchanges its labels. The converse is the argument of Theorem 6.3. ∎

For Fibonacci, with $F$ from Theorem 6.6, the unknowns are $R_1=R_1^{\tau\tau}$ and $R_\tau=R_\tau^{\tau\tau}$. Two instances of the first family, with $a=b=c=\tau$, are enough. With $d=1$ and $e=g=\tau$, every F-symbol is 1 and the sum has only $f=\tau$:

$$R_\tau^2 = R_1.$$

With every label $\tau$, the left side is $R_\tau(-\varphi^{-1})R_\tau$ and the right side $\varphi^{-1/2}\cdot 1\cdot\varphi^{-1/2} + (-\varphi^{-1})R_\tau(-\varphi^{-1})$, so

$$R_\tau^2 + \varphi^{-1}R_\tau + 1 = 0.$$

**Theorem 7.6** (Fibonacci's R-symbols)**.** *The hexagons for Fibonacci have exactly two solutions,*

$$R_1^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5},$$

*and its complex conjugate.*

*Proof.* The roots of $x^2 + \varphi^{-1}x + 1 = 0$ have product 1 and sum $-\varphi^{-1} = 2\cos(3\pi/5)$, so they are $e^{\pm 3\pi i/5}$. Then $R_1 = R_\tau^2 = e^{\pm 6\pi i/5} = e^{\mp 4\pi i/5}$. Both pairs satisfy every instance of both families (`scripts/check-l7-algebra.py`). ∎

Bonderson (p. 82) and Simon (eq. (10.2)) print the first. The picture alone does not choose; the spin of the next section does.

### Exchanging the last two particles of one state

Fix three Fibonacci particles with total charge $\tau$. Use the channel order $1, \tau$ for both the left trees $L_e$ and the right trees $R_f$ of Lecture 6. The bubble calculation of Proposition 6.2 gives

$$L_e^\dagger L_{e'} = \delta_{ee'}\varphi\ \mathrm{id}_\tau, \qquad R_f^\dagger R_{f'} = \delta_{ff'}\varphi\ \mathrm{id}_\tau\,.$$

Thus the printed trees have squared norm $\varphi$. For normalized physical states use the orthonormal vectors

$$\ell_e = \varphi^{-1/2}L_e, \qquad r_f = \varphi^{-1/2}R_f.$$

The common rescaling leaves the F-matrix unchanged. A unit coefficient on the raw tree $L_1$ would instead have squared norm $\varphi$.

Write the coefficients of a state as a column $c_L$ in the basis $\ell_1, \ell_\tau$. The convention of Definition 6.1 has the input tree on a row: $\ell_e = \sum_f F_{ef} r_f$, with the associator suppressed. Consequently the column in the right basis is $c_R = F^T c_L$. For a unitary F-matrix, the inverse column change is $\overline{F}$. This transpose matters in general. For the real symmetric Fibonacci matrix of Theorem 6.6,

$$F^T = F, \qquad \overline{F} = F, \qquad F^{-1} = F.$$

The first exchange $\sigma_1$ acts diagonally on the left basis. The last-two exchange $\sigma_2$ acts diagonally on the right basis. Use the exchange phases selected by $\theta_\tau$ in § 7.5. Set

$$D = \begin{pmatrix} R_1^{\tau\tau} & 0 \\ 0 & R_\tau^{\tau\tau} \end{pmatrix}, \qquad R_1^{\tau\tau} = e^{-4\pi i/5}, \quad R_\tau^{\tau\tau} = e^{3\pi i/5}.$$

To exchange the last two particles, expand in the right basis, apply $D$, then expand back in the left basis, as in Figure 7.1. The two changes of basis preserve the state. Only the crossing is a physical exchange; no intermediate channel is measured.

The column-action matrices are therefore

$$M_1 = D, \qquad M_2 = \overline{F}DF^T = FDF.$$

The last equality uses the displayed Fibonacci gauge. For a general unitary F-matrix the formula is $\overline{F}DF^T$.

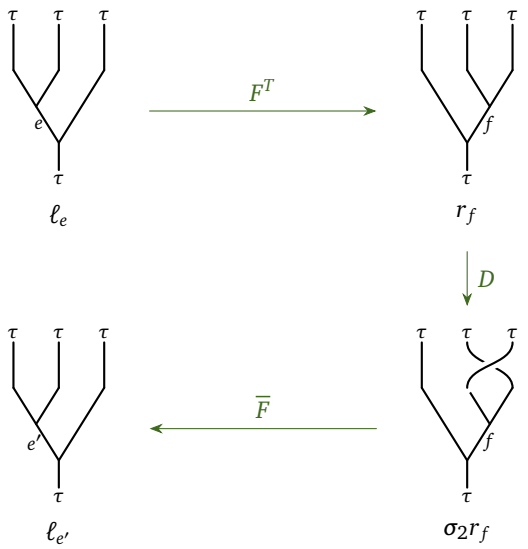



Figure 7.1: The bases used to calculate $\sigma_2$. Every tree includes the common factor $\varphi^{-1/2}$. The arrows give operations on coefficient columns; the crossing uses the left-over-right convention. The channels $e$, $f$ and $e'$ range over $1, \tau$.

Take the normalized initial state $\ell_1$, whose first pair has vacuum charge. Its coefficient column passes through the three steps as follows:

$$\begin{aligned} c_L = \begin{pmatrix} 1 \\ 0 \end{pmatrix} &\longmapsto c_R = \begin{pmatrix} \varphi^{-1} \\ \varphi^{-1/2} \end{pmatrix} \\ &\longmapsto Dc_R = \begin{pmatrix} \varphi^{-1} R_1^{\tau\tau} \\ \varphi^{-1/2} R_\tau^{\tau\tau} \end{pmatrix} \\ &\longmapsto c'_L = \begin{pmatrix} \varphi^{-2} R_1^{\tau\tau} + \varphi^{-1} R_\tau^{\tau\tau} \\ \varphi^{-3/2} (R_1^{\tau\tau} - R_\tau^{\tau\tau}) \end{pmatrix}. \end{aligned}$$

Substituting the two exchange phases gives the output state

$$\sigma_2 \ell_1 = \varphi^{-1} e^{4\pi i/5} \ell_1 + \varphi^{-1/2} e^{-3\pi i/5} \ell_\tau .$$

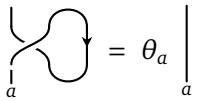
The first-pair charge is no longer definite. A subsequent measurement of that charge has probabilities

$$P(1) = \varphi^{-2}, \qquad P(\tau) = \varphi^{-1}, \qquad P(1) + P(\tau) = 1,$$

because $\varphi^{-2} + \varphi^{-1} = 1$. The relative phase remains part of the coherent state before that measurement. This exchange mixes two amplitudes while preserving the external particle types, the total charge and the norm. Sheet 7 checks the braid relation for $M_1$ and $M_2$.

## 7.5 Topological spin

Closing the right strand of $c_{a,a}$, its partial trace in the sense of (4.9), gives an endomorphism of the simple $a$, hence a number, the *topological spin* $\theta_a$:




(7.7)

**Theorem 7.7** (The spin from R · *proved given coherence*)**.** *In a unitary multiplicity-free fusion category with its canonical spherical structure,*

$$\theta_a = \frac{1}{d_a}\sum_c N_{aa}^{c}\, d_c\, R_c^{aa}.$$

*Proof.* Take the trace of both sides of (7.7): the left side becomes the trace of $c_{a,a}$, and the right side $\theta_a d_a$. On each channel $c$ of $a\otimes a$, $c_{a,a}$ acts as $R_c^{aa}$ by (7.5), so it is $\sum_c R_c^{aa}P_c$ with $P_c$ the projections of (5.3), and the trace of $P_c$ is $N_{aa}^{c}d_c$. ∎

For Fibonacci,

$$\theta_\tau = \frac{e^{-4\pi i/5} + \varphi\, e^{3\pi i/5}}{\varphi} = e^{4\pi i/5},$$

and the conjugate solution gives $e^{-4\pi i/5}$. The course takes $\theta_\tau = e^{4\pi i/5}$, which fixes the chirality. The spin is a gauge invariant, like the double braiding: a gauge changes $R_c^{ab}$ by $u_c^{ba}/u_c^{ab}$ and $R_c^{ba}$ by the inverse. The two are related by the *ribbon relation* $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$ (Bonderson (2.60) · *cited*); Lecture 9 identifies $\theta_a$ with the twist of a ribbon category, the phase of rotating one particle by $2\pi$.

## 7.6 Pointed braidings; Ising

For $\mathrm{Vec}_G^{\omega}$ on $\mathbb{Z}_2 = \{1, s\}$, a braiding is determined by one number, written $c(s,s) = R_1^{ss}$. By Theorem 7.7, $\theta_s = c(s,s)$.

**Theorem 7.8** (Braidings on $\mathbb{Z}_2$)**.** *The hexagons for* $\mathrm{Vec}_G^{\omega}$ *on* $\mathbb{Z}_2$ *force* $c(s,s)^2 = \omega(s,s,s)$*, and each such* $c(s,s)$ *gives a braiding* · *sketched. So the trivial* $\omega$ *has two braidings,* $c(s,s) = \pm 1$*, the* boson *and the* fermion *(*sVec*); and* $\omega(s,s,s) = -1$ *has two,* $c(s,s) = \pm i$*, the* semion *and the* antisemion*.*

*Proof.* In the first family take $a = b = c = d = s$ and $e = g = 1$, so that $f = 1$: the left side is $c(s,s)\,\omega(s,s,s)\,c(s,s)$ and the right side $\omega(s,s,s)\cdot 1\cdot\omega(s,s,s)$. So $c(s,s)^2 = \omega(s,s,s)$. Conversely every instance of both families holds when it does (`scripts/check-l7-algebra.py`), and the converse of Theorem 7.5 gives the braiding. ∎

Here $\omega$ was fixed and the braiding varied. A *quadratic form* on a finite abelian group $A$ is a function $q\colon A \to \mathbb{C}^\times$ with

$$q(-x) = q(x) \qquad (x \in A),$$

for which the ratio

$$\frac{q(x+y)}{q(x)q(y)} \qquad (x, y \in A)$$

is multiplicative in each argument. Pointed braided fusion categories are classified, up to braided equivalence, by these forms up to automorphisms of $A$ · *cited*. Every form occurs, with $q(x) = \theta_x$, and determines $\omega$ up to a coboundary. For $\mathbb{Z}_5$, the five possible functions $q(x) = e^{2\pi i k x^2/5}$ give three classes: $k = 0$, $k \in \{1,4\}$ and $k \in \{2,3\}$ (EGNO Thms. 8.4.9 and 8.4.12, after Eilenberg–Mac Lane and Joyal–Street).

**Theorem 7.9** (Braidings on Ising · *cited*)**.** *Each of the two Ising fusion categories has exactly four braidings (Siehler, Thm. 1.2). On the course's Ising* ($\kappa = +1$) *they have* $\theta_\sigma = e^{i\pi\nu/8}$ *with* $\nu \in \{1,7,9,15\}$*; the other odd* $\nu$ *belong to the* $\kappa = -1$ *category, which contains* $\mathrm{SU}(2)_2$ *(*$\nu = 3$*), and* $\kappa_\sigma = (-1)^{(\nu^2-1)/8}$ *(Kitaev, eqs. (91)–(96) and Table 1).*

The course's Ising is $\nu = 1$; in Lecture 6's gauge (Kitaev's Table 1, Bonderson p. 84),

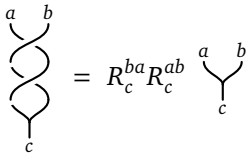
$$R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{3i\pi/8},$$
$$R_\sigma^{\sigma\psi} = R_\sigma^{\psi\sigma} = -i, \quad R_1^{\psi\psi} = -1.$$

With the $F$ of Lecture 6 these satisfy both hexagon families (the script checks them; Sheet 7 solves them). The spin formula gives

$$\theta_\sigma = \frac{e^{-i\pi/8} + e^{3i\pi/8}}{\sqrt{2}} = e^{i\pi/8}\,\frac{2\cos(\pi/4)}{\sqrt{2}} = e^{i\pi/8}.$$

The spin of $\psi$ is $\theta_\psi = -1$, so $\psi$ is a fermion.

On a vertex the double braiding is a number, since both crossings act by R-symbols:




(7.8)

## Toric-code data

The *toric code* is an anyon model with four particle types $1, e, m, \varepsilon$. Its fusion is the group law of $\mathbb{Z}_2 \times \mathbb{Z}_2$. The quantum dimensions, topological spins and nontrivial products are

| $a$ | $1$ | $e$ | $m$ | $\varepsilon$ |
|---|---|---|---|---|
| $d_a$ | 1 | 1 | 1 | 1 |
| $\theta_a$ | 1 | 1 | 1 | $-1$ |

$$e \otimes e = m \otimes m = \varepsilon \otimes \varepsilon = 1,$$
$$e \otimes m = \varepsilon, \quad e \otimes \varepsilon = m, \quad m \otimes \varepsilon = e.$$

Here the fusion equalities identify simple objects up to isomorphism. The rule $e \otimes m = \varepsilon$ identifies $\varepsilon$, the only fermion. These data do not distinguish $e$ from $m$. Exchanging them is a symmetry (Lecture 12); the lattice convention in Lecture 13 fixes which is which. The double braiding of $e$ with $m$ is $-1$. It is the phase for taking $e$ once around $m$; one exchange alone need not have this phase in every gauge. Sheet 7 constructs this data from a bicharacter, meaning a scalar braiding that is multiplicative in each argument. Lecture 12 recovers the same model as a centre.

Finally, an object $X$ that braids trivially with everything, $c_{Y,X}c_{X,Y} = \mathrm{id}$ for all $Y$, is *transparent*; the transparent objects form the *Müger centre* (EGNO's centralizer $C'$, §8.20). In sVec every object is transparent. In Fibonacci the only transparent simple object is 1, since $R_c^{\tau\tau}R_c^{\tau\tau} \neq 1$ for both channels. A *trivial* Müger centre consists of direct sums of the unit. Lecture 10 shows that this holds exactly when the S-matrix is invertible.

**Summary.**

- *Handed on:* in a plane, exchanges form the braid group $B_n$ (Theorem 7.1); a braiding is a natural $c_{X,Y}$ with two hexagons (Definition 7.2), and it gives a representation of $B_n$ (Theorem 7.4); Yang–Baxter in braid form is Reidemeister III.
- *R-symbols:* $c_{a,b}\psi_c^{ab} = R_c^{ab}\psi_c^{ba}$, constrained by the hexagon equations in F and R (Theorem 7.5); for Fibonacci $R_1^{\tau\tau} = e^{-4\pi i/5}$, $R_\tau^{\tau\tau} = e^{3\pi i/5}$ or the conjugate (Theorem 7.6).
- *Spin:* $\theta_a = d_a^{-1}\sum_c N_{aa}^c d_c R_c^{aa}$, the value of the closed curl (Theorem 7.7); $\theta_\tau = e^{4\pi i/5}$.

- *Pointed and Ising:* on $\mathbb{Z}_2$, $c(s,s)^2 = \omega(s,s,s)$: boson, fermion, semion, antisemion (Theorem 7.8); Ising has four braidings, and the course's has $\theta_\sigma = e^{i\pi/8}$ (Theorem 7.9).
- *Next:* Lecture 8 builds braidings from Hopf algebras: quantum groups.

## References

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